

Cloud Networking

Session 1 and Session 2

Name	<i>Piyush Kumar</i>
Employee ID	<i>513558</i>
Team / Department	<i>Cloud and Data Center</i>
Mentor	<i>Bharathraj B R</i>
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1. Introduction

This document presents a structured summary of the learning outcomes from two cloud networking training sessions conducted as part of Autodesk's internal technical development program. The sessions provided an in-depth introduction to Autodesk's cloud network architecture, the AWS Network Manager platform, global topology visualization, corporate connectivity solutions, and foundational core network concepts.

The content has been documented from the perspective of a training participant and reflects personal understanding and interpretation of the material covered. All acronyms are expanded upon first use where the full form is confidently known. Technical details are presented at a conceptual level unless specific data was explicitly provided during the sessions.

This summary is intended for submission to a manager, trainer, or internal technical team and is suitable for review as evidence of session participation and knowledge acquisition.

2. Cloud Environments and Usage Categories

The sessions began with an overview of the three distinct cloud environments maintained by Autodesk. Understanding the purpose and access model of each environment is fundamental to navigating Autodesk's cloud infrastructure. Each environment serves a specific user base and operational requirement, and they operate under different security and access controls.

2.1 Environment Overview

Corporate Environment:

The Corporate environment is designed exclusively for Autodesk employees and internal engineering teams. It supports internal development and operational activities, including work with GitHub repositories, source code, development pipelines, and internal tooling. Access is restricted to internal users and is governed by Autodesk's corporate identity and access management policies.

Commercial Environment:

The Commercial environment hosts customer-facing products and services. This includes flagship Autodesk applications such as Revit and AutoCAD. Services within this environment may be made accessible to eligible external customers and end users, depending on product-level permissions and the applicable access controls. The commercial environment is built for scale and reliability to support Autodesk's global customer base.

FedRAMP Environment:

The FedRAMP (Federal Risk and Authorization Management Program) environment is a highly controlled and restricted operating environment designed to meet United States government security and compliance requirements. Access and operations within this environment are subject to strict authorization controls and regulatory constraints. It is important to note that not every FedRAMP system is directly operated by the US government; rather, this environment is architected and maintained to satisfy the compliance mandates applicable to government-aligned workloads.

2.2 Environment Comparison

The following table provides a side-by-side comparison of the three environments:

Environment	Primary Users	Main Purpose	General Access Level
Corporate	Autodesk employees and internal teams	Engineering, development, source code repositories, GitHub, internal tooling	Restricted to internal users
Commercial	External customers and end users	Customer-facing products such as Revit and AutoCAD	Varies by product and access controls
FedRAMP	US government-aligned stakeholders	Applications meeting US government security and compliance requirements	Highly restricted; subject to strict authorization

3. Autodesk Corporate AWS Network Manager

3.1 Global Network Overview

AWS Network Manager is a centralized network management service that provides a unified view of an organization's global network infrastructure on AWS. Within the context of these training sessions, it was used to explore and visualize Autodesk's corporate global network. The global network reviewed during the sessions comprised the following high-level components:

- 4 edge locations distributed across AWS regions
- 10 segments used to logically separate network traffic
- 2 Network Function Groups (NFGs) supporting network policy enforcement and routing functions

3.2 Edge Location Details

The following table captures the edge location data presented during the sessions. Each edge location represents a regional point of presence within Autodesk's global corporate network, identified by its AWS region. The table records the number of Virtual Private Clouds (VPCs) and Transit Gateways (TGWs) associated with each region as reviewed in AWS Network Manager:

AWS Region	Number of VPCs	Number of TGWs
us-east-1	699	5
ap-southeast-1	14	3
us-west-2	739	6
eu-west-1	79	2

3.3 Key Terminology

The following terms were introduced and explained during the sessions:

VPC (Virtual Private Cloud)	A logically isolated network environment within AWS that enables organizations to define and control their own virtual networking components, including subnets, routing tables, and gateways.
TGW (Transit Gateway)	A highly scalable AWS networking construct that acts as a central hub to interconnect VPCs, on-premises networks, and other AWS services. It simplifies complex network architectures by enabling hub-and-spoke connectivity.
Edge Location	In the context of Autodesk's global corporate network, an edge location represents a regional network boundary where Autodesk's corporate network connects to AWS infrastructure, typically anchored by a Transit Gateway.
Segment	A logical partition of the core network used to isolate and control traffic flows between different groups of network resources. Segments enforce network segmentation policies and facilitate controlled inter-segment communication.
Network Function Group (NFG)	A grouping construct within AWS Network Manager used to associate network functions — such as security appliances or inspection services — with specific traffic flows or segments.

4. Global Network Topology Exploration

A significant portion of the sessions was dedicated to hands-on exploration of Autodesk's global corporate network topology within AWS Network Manager. Two primary views were examined: the topology graph and the topology tree. These views provide complementary perspectives on the same underlying network infrastructure.

4.1 Topology Graph

The topology graph provides a visual, node-and-edge representation of network resources and their interconnections. It displays resources as graphical nodes and the relationships between them as connecting lines or edges. This view is particularly effective for understanding the overall shape of the network and identifying how components are interconnected at a glance. Engineers can use the topology graph to assess network layout, trace connectivity between specific resources, and identify potential points of congestion or failure.

4.2 Topology Tree

The topology tree provides a hierarchical, structured view of the global network and its components. Resources are arranged in a parent-child relationship that reflects their logical or organizational hierarchy within the network. This view is useful for systematically browsing nested resources and understanding the containment relationships between network constructs — for example, how VPCs are nested within a Transit Gateway or how segments relate to the core network.

4.3 Network Resource Components

During the topology exploration, several resource types were encountered in both the graph and tree views. The following components were identified and reviewed:

CNE	CNE was encountered as a component type within the topology views. The full form of this acronym was not confirmed during the sessions and is therefore retained as-is. It was reviewed as part of the global network topology.
VPC (Virtual Private Cloud)	Isolated network environments within AWS regions that form the foundational building blocks of Autodesk's cloud network. Multiple VPCs were visible across different regions in both topology views.
Connect	A connectivity construct visible in the topology that facilitates specific types of network attachment or peering, enabling resources to integrate with the core network.
Segment	Logical network partitions visible in the topology, used to enforce separation and control traffic flows between network groups.
Device	Physical or virtual network devices registered in AWS Network Manager, representing infrastructure elements such as routers or on-premises appliances.
VPN (Virtual Private Network)	Encrypted tunnel connections visible in the topology, used to securely extend on-premises or remote networks into the AWS environment.
Direct Connect	Dedicated private network connections between Autodesk's on-premises infrastructure and AWS, bypassing the public internet for improved performance and security.

4.4 Value of Topology Visualization

The topology visualization capabilities explored during the sessions offer meaningful operational value for network engineers. Specifically, these views support the following activities:

- Understanding network relationships: The graphical representation makes it easier to comprehend how disparate components are connected and dependent on one another.
- Identifying connected resources: Engineers can trace connectivity chains, allowing them to identify all resources attached to a particular hub, segment, or gateway.
- Reviewing regional connectivity: The global view within AWS Network Manager allows review of cross-region connections and the distribution of resources across edge locations.
- Supporting troubleshooting and network analysis: By visualizing the actual network state, topology tools enable faster root-cause identification during network incidents and facilitate pre-change impact assessments.

5. Corporate Network Technologies and Solutions

The sessions introduced several technologies and vendor solutions that form part of Autodesk's corporate network architecture. The descriptions below represent conceptual-level explanations as covered during the sessions. Autodesk-specific implementation details were not disclosed.

5.1 SASE Solution

Secure Access Service Edge (SASE) is a network architecture model that converges wide-area networking (WAN) capabilities with comprehensive cloud-delivered security functions into a single, unified service framework. Rather than routing traffic through a central data centre for inspection and policy enforcement,

SASE enables security policies to be applied at the network edge — closer to users, devices, and applications — regardless of their physical location.

Key characteristics of SASE include identity-driven access control, zero-trust security principles, integrated threat protection, and support for cloud-native and distributed workforces. By combining networking and security into a unified cloud service, SASE reduces complexity and improves the performance and security posture of enterprise networks. This approach is particularly relevant for organizations like Autodesk, where employees and systems operate across multiple regions and cloud environments.

5.2 Palo Alto Networks

Palo Alto Networks is a leading provider of enterprise cybersecurity solutions, with a strong focus on network security and firewall technology. During the sessions, Palo Alto was referenced in the context of Autodesk's corporate network security infrastructure. At a general level, Palo Alto's products include next-generation firewalls (NGFWs), cloud-delivered security services, and threat intelligence capabilities.

In enterprise network environments, Palo Alto solutions are commonly deployed to enforce security policies, inspect network traffic for threats, and control access between network segments. The specific implementation details of Palo Alto's role within Autodesk's network were not discussed in these sessions; however, its inclusion in the curriculum indicates its relevance to Autodesk's network security architecture.

5.3 AWS Cloud WAN

AWS Cloud WAN (Wide Area Network) is a managed service that enables organizations to centrally build, manage, and monitor a global wide-area network using AWS's infrastructure. It allows network engineers to define network policies through a central dashboard and automatically provisions the necessary AWS resources — such as Transit Gateways and core network edges — across multiple regions.

Cloud WAN simplifies the process of building and maintaining complex multi-region network architectures by abstracting the underlying infrastructure. It integrates natively with AWS Network Manager, providing a consolidated view of the global network alongside policy management capabilities. For organizations with workloads spanning multiple AWS regions and on-premises sites, Cloud WAN offers a scalable and operationally efficient connectivity solution.

6. Core Network Concepts

The sessions also covered three foundational concepts relevant to the AWS Cloud WAN core network architecture. These concepts describe how the core network is configured and maintained at a structural level. The explanations below are conceptual in nature; Autodesk-specific configuration details were not provided.

6.1 Attachments

In the context of AWS Cloud WAN, attachments are the connections that associate network resources with the core network. They represent the integration points between the core network and the components that participate in it — such as VPCs, Transit Gateways, VPN connections, and Direct Connect gateways. Each attachment defines how a resource is connected to the core network and to which segment it belongs. Managing attachments is a primary mechanism through which network engineers control which resources can communicate with one another and under what policy rules.

6.2 Peerings

Peerings describe connectivity relationships established between supported network constructs — most commonly between Transit Gateways or between a Transit Gateway and the core network. Peering allows traffic to flow between two otherwise independent network domains without traversing the public internet. In the context of Cloud WAN, Transit Gateway peering enables the extension of the core network's reach to regions or environments where a core network edge may not be directly deployed. Peerings are an important tool for building scalable and resilient global network topologies.

6.3 Policy Versions

Policy versions represent distinct, saved iterations of the core network configuration policy in AWS Cloud WAN. The core network policy is a document that defines the network's structure — including segments, segment actions, attachment policies, and routing configurations. As network requirements evolve, new policy versions can be created, reviewed, and applied. Maintaining version history provides an audit trail of configuration changes and supports rollback capabilities if an updated policy introduces unintended behaviour. Policy version management is a critical practice for maintaining network consistency and governance in a large-scale cloud WAN environment.

7. Key Takeaways

The following points summarize the most significant learning outcomes from the two cloud networking training sessions:

- Gained a clear understanding of the three Autodesk cloud environments — Corporate, Commercial, and FedRAMP — their distinct purposes, user bases, and access controls.
- Learned how Autodesk's global corporate network is represented and managed within AWS Network Manager, including the structure of edge locations, VPCs, Transit Gateways, segments, and Network Function Groups.
- Explored the topology graph and topology tree views in AWS Network Manager, developing familiarity with visual network exploration and resource relationship mapping.
- Reviewed key network resource component types, including VPC, Connect, Segment, Device, VPN, Direct Connect, and CNE (full form retained as reviewed during sessions).
- Developed an introductory understanding of the SASE architecture model, its convergence of networking and security, and its applicability to distributed enterprise environments.
- Gained awareness of Palo Alto Networks as a network security solution provider relevant to Autodesk's corporate infrastructure.
- Understood AWS Cloud WAN as a managed global WAN service that simplifies multi-region network management through centralized policy definition and automated provisioning.
- Learned the conceptual roles of core network components — attachments, peerings, and policy versions — in the AWS Cloud WAN architecture.

8. Conclusion

These two training sessions provided a comprehensive foundational introduction to Autodesk's cloud networking landscape. Beginning with an understanding of the corporate, commercial, and FedRAMP

environments, the sessions progressively built toward a practical familiarity with how Autodesk's global network is structured and visualized within AWS Network Manager.

The exploration of topology graph and topology tree views equipped attendees with the ability to navigate and interpret the global network visually — a skill directly applicable to day-to-day network engineering and troubleshooting tasks. The introduction of corporate connectivity technologies — SASE, Palo Alto Networks, and AWS Cloud WAN — provided important context for understanding the security and connectivity architecture underpinning Autodesk's cloud operations.

Finally, the review of core network concepts — attachments, peerings, and policy versions — established the conceptual vocabulary necessary to engage with AWS Cloud WAN's configuration and governance model. Collectively, these sessions have provided a solid foundation upon which deeper specialization in Autodesk's cloud networking infrastructure can be developed.